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RESEARCH ARTICLE

Predator decision-making shapes the dynamics and stability of mimicry systems

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Abstract

1. Mimicry is an anti-predator strategy in which prey species (the mimic) resemble an unprofitable species (the model) to deceive predators. Despite theoretical expectations for perfect mimicry, imperfect mimicry, where the mimic resembles its model imperfectly, is widespread in nature.
2. To understand how imperfect mimicry can persist ecologically, we studied the effect of different predator recognition processes on the dynamics and stability of various mimicry systems. Specifically, we extended a dynamical model that integrates optimal foraging and signal detection theories by introducing a novel abundance-dependent recognition mechanism, where predators' perception of the similarity between mimic and model is influenced by the relative abundance of prey types.
3. We demonstrate that intermediate similarity promotes stable community dynamics and increases mimic abundance in single Batesian mimicry systems. Moreover, abundance-dependent recognition leads predators to reduce attack on mimics with low morphological similarity, further contributing to system stability.
4. Extending the framework to a multi-mimicry system, we find that Batesian and Müllerian mimics have contrasting effects: Intermediate Batesian similarity continues to stabilize the system, while high Müllerian similarity provides additional protection and can offset destabilization caused by highly similar Batesian mimics.
5. Our study offers a novel explanation for the prevalence of imperfect mimicry in nature and highlights how recognition processes shape the ecological stability of mimicry systems.

KEYWORDS
Batesian mimicry, imperfect mimicry, mimicry complex, morphological similarity, Müllerian mimicry, optimal foraging theory, signal detection theory

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1 | INTRODUCTION

Mimicry is a complex anti-predator adaptation that involves interactions among three main players: the mimic, the model and the predator (Quicke, 2017; Ruxton et al., 2004). In this system, the model is an unprofitable prey that displays aposematic signals to advertise its unprofitability. The mimic, on the contrary, displays phenotypes that resemble those of the model, including behaviour, chemical compounds, and, more commonly, morphological characteristics, such as colour, pattern and shape (Kikuchi et al., 2013). These phenotypes deceive predators into misidentifying the mimic as the unprofitable model. One can categorize a mimicry system based on the mimic's profitability to the predator, with Batesian and Müllerian mimicry being the two most widely recognized forms. In Batesian mimicry, the mimic is a profitable prey that reduces its predation risk by resembling an unprofitable model. However, the model is predicted to suffer higher mortality as the presence of the Batesian mimic confuses the predators, thereby reducing the effectiveness of the model's aposematic display (Bates, 1862, though empirical studies report mixed outcomes; Akcali et al., 2018; Lindström et al., 2004). In Müllerian mimicry, the mimic is an unprofitable prey that shares a common aposematic signal with the model. Unlike Batesian mimicry, both the mimic and the model benefit from their resemblance in Müllerian mimicry due to the positive reinforcement of the aposematic signal and by receiving fewer per capita attacks (Mallet & Joron, 1999; Müller, 1879; Ruxton et al., 2004; Sherratt, 2008; Vane-wright, 1980). It has been suggested that natural selection should favour mimics that closely resemble their model in both mimicry systems, resulting in highly similar mimic–model pairs (Nur, 1970; Ruxton et al., 2018; Sherratt, 2002). However, empirical evidence shows that many mimics do not closely resemble their models despite the apparent advantage of high similarity, a phenomenon known as imperfect mimicry (Bosque et al., 2018; Kikuchi et al., 2013; McLean et al., 2019; Sherratt, 2002; Sherratt & Peet-Paré, 2017).

This contradictory phenomenon raises the question: what mechanisms allow imperfect mimicry to persist in nature? Many hypotheses and theories have been proposed to explain the existence of imperfect mimicry (Kikuchi et al., 2013; Penney et al., 2012; Pfennig & Kikuchi, 2012). On the one hand, evolutionary explanations, such as the 'chase-away' hypothesis, predict that imperfect mimicry exists because models evolve away from their mimic (Akcali et al., 2018; Franks et al., 2009; McGuire et al., 2006; Nur, 1970; Oaten et al., 1975). On the other hand, ecological explanations, such as the 'eye of the beholder' hypothesis, argue that what is perceived as imperfect mimicry by humans may in fact be effective from the perspective of the predator (Cuthill & Bennett, 1993; Dittrich et al., 1993). Furthermore, the 'multiple models and multiple predators' hypothesis suggests that imperfect mimicry could arise either because the mimic resembles multiple models or because multiple predators rely on different signals to detect the mimic (Edmunds, 2000; Pekár et al., 2011; Sherratt, 2002). As a result, the mimic might resemble many aposematic signals from different models, resulting in a signal that partially resembles several model species but does not closely match any single one (also see Kikuchi et al., 2013 for additional ecological hypotheses).

In addition to morphological similarities, different types of predator recognition processes could also influence the dynamics of mimicry systems (Chittka & Osorio, 2007; Darst, 2006). These recognition processes shape the realized similarity perceived by the predator, which may differ from the actual mimic–model morphological similarity. Predator recognition can be affected by the learning efficiency of the predator (Huheey, 1964) or by the cost of learning (Kikuchi & Sherratt, 2015). Empirical work also suggests that predator recognition can depend on relative prey abundance; for example, predators exposed to a higher abundance of toxic prey learn avoidance more effectively and subsequently alter their attack behaviour (Nelson et al., 2011). Müllerian mimicry is another classic example of such abundance-dependent recognition—a higher abundance of unprofitable mimics increases predator encounters and reinforces the association between the aposematic signal and unprofitability (Aubier et al., 2017; Müller, 1879). This type of abundance-dependent recognition is particularly important on an ecological timescale as it creates intricate feedback between prey abundance, predator recognition, predator attack decisions and subsequent community dynamics.

Despite numerous hypotheses being proposed to address the emergence of imperfect mimicry, its effects on population dynamics and the stability of mimicry systems remain unclear. Understanding these ecological mechanisms and their consequences clarifies how imperfect mimicry persists over short ecological timescales, which is essential for setting the stage for its evolution over longer timescales. Previous theoretical studies addressing the stability of mimicry systems often lacked comprehensive dynamics of the entire system: some overlooked the dynamics of the predator (e.g. Getty, 1985), while others omitted both the predator and the non-mimetic alternative prey (e.g. Kumazawa et al., 2006; Yamauchi, 1993). Other studies have focused on how mimicry dynamics are shaped by factors such as the influx of alternative prey, the degree of similarity between the mimic and the model, and the level of defence possessed by the model species (Brower & Moffitt, 1974; Kikuchi et al., 2022; Rowell-Rahier et al., 1995). Many of these theoretical studies employed optimal foraging theory, which assumes that predators make attack decisions based on properties of the prey items (i.e. their profitability and abundance; Charnov, 1973, 1976; Stephens & Krebs, 1986).

Although previous theoretical studies on mimicry population dynamics have mostly focused on a single Batesian mimic system based on morphological similarity, natural mimicry often forms multi-mimicry complexes involving multiple types of mimicries and various recognition processes. Specifically, multi-mimicry complexes involving both Batesian mimicry and Müllerian mimicry are common in nature (e.g. *Heliconius* butterflies, Quicke, 2017; *Pachyrhynchus* weevils, Schultze, 1923), where abundance-dependent recognition may also play an important role. With a greater number of prey types involved, the predator needs to juggle between multiple mimic–model similarities and prey profitabilities when making attack decisions. Importantly, Müllerian mimics differ fundamentally from Batesian mimics: the resemblance between the Müllerian mimic and the model produces a positive

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reinforcement that provides both species greater protection, whereas the resemblance between the Batesian mimic and the model imposes a cost on the model species. Therefore, Batesian and Müllerian mimics may differentially influence the dynamics of the multi-mimicry system. To investigate the persistence of imperfect mimicry in natural systems with both Batesian mimicry and Müllerian mimicry, it is essential to consider the interactive effect of their similarity with the model and their profitability for the predator.

Here, we extended the framework proposed by Kikuchi et al. (2022), which offers an exciting opportunity to investigate the persistence of imperfect mimicry and its resulting community dynamics by integrating optimal foraging theory into a dynamical model of the full mimicry system. By exploring the impact of mimic–model similarity in a single Batesian mimicry system, we offer a novel perspective on the ecological dynamics of mimicry systems: imperfect mimicry can stabilize community dynamics and lead to a higher mimic abundance. We then explore how this stability pattern conferred by imperfect mimicry varied with different predator recognition mechanisms. To this end, we formulated a novel phenomenological representation of the predator's recognition process, which depends on both the innate mimic–model similarity (i.e. morphological similarity based on prey traits) and the abundance ratio of the mimic and the model. Finally, we expanded the framework to a multi-mimicry system encompassing both Batesian and Müllerian mimicry. We explored how the dynamics of the multi-mimicry system are influenced by the similarities between different mimics and the model under the aforementioned abundance-dependent recognition-determining process. Overall, we show that imperfect mimicry can, counterintuitively, promote higher mimic abundance and greater stability across a range of mimicry systems that differ in predator recognition mechanism and the number of mimic species involved.

2 | METHOD

We used a theoretical model to study how mimic–model similarity and different predator recognition processes influence predation decisions and the dynamics of mimicry systems. Our ordinary differential equation (ODE) model is built upon the theoretical framework of Kikuchi et al. (2022), which combines optimal foraging theory and signal detection theory to simulate the dynamics of mimicry systems. We extended this framework to investigate how mimic–model similarity and predators' abundance-dependent recognition influence the abundance of mimic species and the stability of the system through predator decision-making (Figure 1). In the following sections, we first introduce the foundations determining the predator decision-making process, including optimal foraging theory, signal detection theory and abundance-dependent recognition. We then present the population dynamic framework that governs the abundance of species within the mimicry system. Finally, we describe the numerical setup used in our study.

2.1 | Predator decision-making process

2.1.1 | Optimal foraging theory

We followed the classic assumption that predators determine their probability (p_i) of attacking three different types of prey items—mimics (N_{c_j}; $j = 1, 2, \cdots, n$), model (N_m) and alternative prey (N_n)—through optimal foraging theory (Charnov, 1973; Charnov, 1976; Figure 1a). Assuming a multi-prey Holling type-II functional response, the energy gain term that predators attempt to maximize while making foraging decisions is:

$$\text{Predator energy gain} = \frac{s\left(\sum_j N_{c_j} p_{c_j} v_{c_j} + N_m p_m v_m + N_n p_n v_n\right)}{1 + s\left(\sum_j N_{c_j} p_{c_j} h_{c_j} + N_m p_m h_m + N_n p_n h_n\right)}. \quad (1)$$

Here, p_i represents the probability of the predator attacking prey item i ($i = c_j, m, n$ for mimics, model and alternative prey, respectively). The parameter v_i represents the value that the predator gains from consuming one individual of prey item i, whereas h_i represents the time predators spend handling one individual of prey item i. Therefore, each $N_i p_i v_i$ term in the numerator represents the value gain of consuming species i, whereas each $N_i p_i h_i$ term in the denominator represents the cost of consuming species i. The parameter s represents the predator's search rate, which is assumed to be equal for all prey items. Here, we defined a prey item's profitability based on its $\frac{v_i}{h_i}$ (Charnov, 1976). We further assumed that at least one mimic species is a Batesian mimic and will have the highest profitability, followed by the alternative prey, and, finally, the model. This ordering ensures that mimicry creates a strong trade-off for the predator between energetic gain and mistaken consumption of the model. Therefore, when there is only one mimic species (i.e. the Batesian mimic; $j = 1$), this parameterization ensures that $\frac{v_{c_1}}{h_{c_1}} > \frac{v_n}{h_n} > \frac{v_m}{h_m}$ (i.e. the Batesian mimic will be the most profitable prey). In classical optimal foraging theory, which assumes that predators have complete information about the system and can instantaneously recognize different prey types, predator behaviour always follows an all-or-nothing manner (Charnov, 1976; Stephens & Krebs, 1986). That is, predators will either always include that prey in their diet ($p_i = 1$) or never include that prey in their diet ($p_i = 0$) upon encounter, with a decision-switching threshold determined by the profitability and abundance of prey items (Stephens & Krebs, 1986).

2.1.2 | Signal detection theory

In mimicry systems, however, predators may experience difficulty in distinguishing between the mimic and the model. Such imperfect information results in a non-binary consumption probability due to the numerical connection between p_{c_j} and p_m. Mathematically, this means

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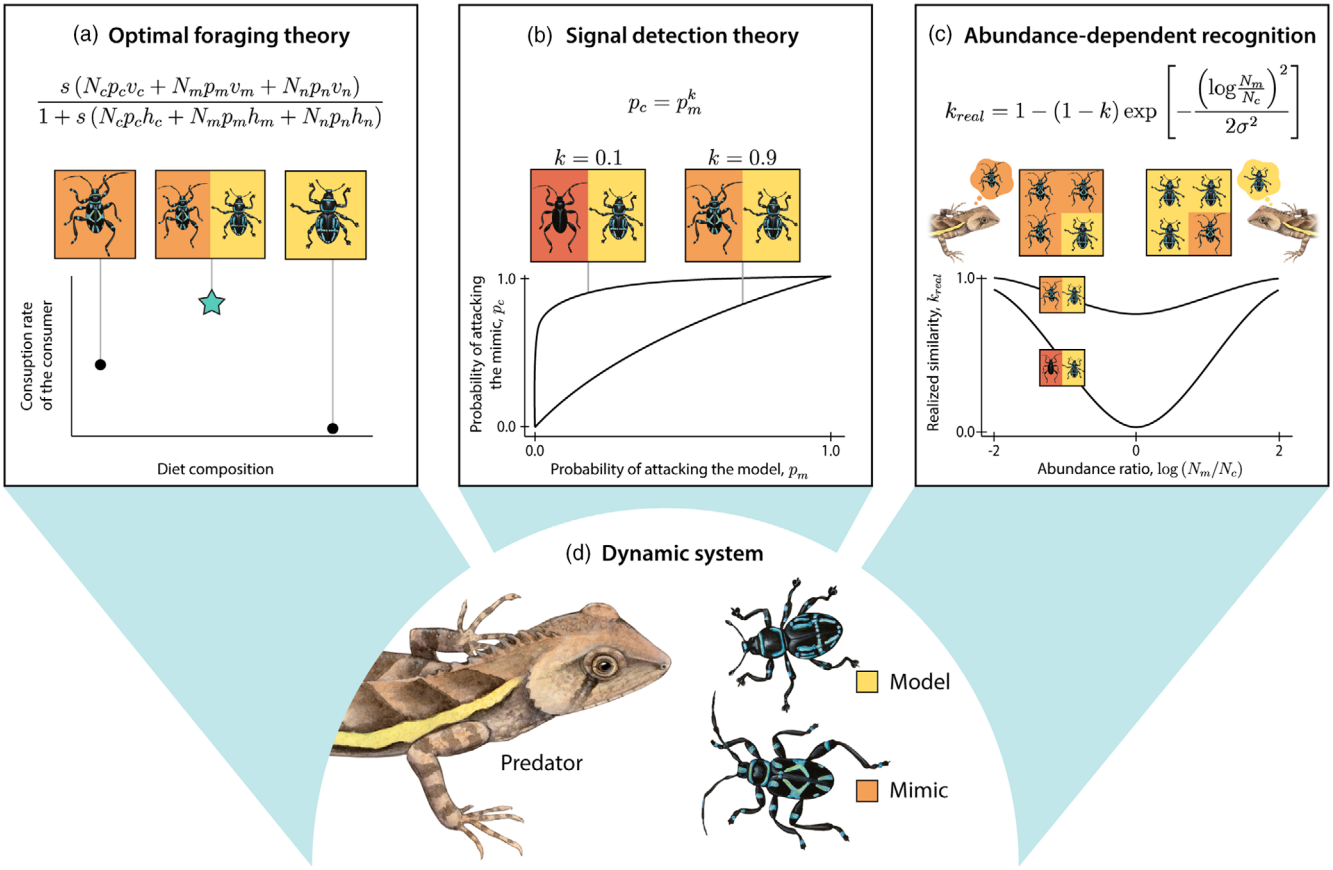


FIGURE 1 Illustrative figure demonstrating how our study incorporates (a) optimal foraging theory (i.e. predator optimizing attack probability p_i to maximize foraging gain), (b) signal detection theory (i.e. non-independence of attack probabilities via mimic–model morphological similarity k), and (c) abundance-dependent recognition (i.e. predator realized recognition k_{real} also dependent of model–mimic abundance ratio) into a (d) dynamic system of mimic–model–predator. Here, only a single Batesian mimic is shown for simplicity (so the secondary subscript j in N_{c_j} is omitted). In panels (b) and (c), we depict two scenarios with low ($k = 0.1$; red mimic) and high ($k = 0.9$; orange mimic) innate mimic–model morphological similarity k. See the main text for the mathematical description of the full multi-mimic model. This figure is illustrated by Julie Johnson from Life Science Studios.

that the predator cannot make independent decisions on p_{c_j} and p_m to maximize Equation 1 (Getty, 1985; Kikuchi et al., 2022). One can first assume that the predator has a fixed recognition ability, which, in the simplest case, depends solely on the morphological similarity between the mimic and the model. Following Getty (1985), the probabilities of attacking the mimic and the model are linked by a power law function:

$$p_{c_j} = p_m^{k_j}, \tag{2}$$

with the exponent $k_j \in [0, 1]$ representing the morphological similarity between mimic N_{c_j} and the model N_m (Figure 1b). When $k_j = 0$, the predator can distinguish between the mimic and the model perfectly, indicating complete discriminability from the predator's perspective. Following classical optimal foraging, in a single Batesian mimic system (i.e. $j = 1$), the predator will always attack the mimic ($p_{c_1} = 1$) due to its higher profitability (i.e. $\frac{v_{c_1}}{h_{c_1}} > \frac{v_m}{h_m}$). On the other end of the spectrum, the two prey items are morphologically identical from the predator's perspective when $k_j = 1$ (i.e. perfect mimicry); the predator would treat the mimic and the model as the same species and either attack or reject them altogether ($p_{c_j} = p_m = 0$ or 1). However, in most natural cases, mimicry falls between the two extreme cases, and k_j will be a number between 0 and 1. Here, $0 < k_j < 1$ represents the scenario when imperfect information interferes with optimal foraging decision-making, leading to non-binary optimal foraging decisions (i.e. $0 < p_{c_j}, p_m < 1$). As k_j approaches one, predators experience greater difficulty distinguishing the mimic from the model, leading to a higher likelihood of attacking the wrong prey (i.e. a higher perceived mimic–model similarity). For our purpose, we defined intermediate similarity as $0.5 < k_j < 0.8$. At the same time, the attack probability of the alternative prey remains consistent with classical optimal foraging theory (i.e. $p_n = 0$ or 1). When extending the framework to multi-mimic systems, each mimic is characterized only by its similarity to the model, and the pairwise resemblance among mimics is not represented separately. This approach assumes that similarity k_j reflects a projection of multidimensional trait dissimilarity onto a single axis, with the model serving as the reference point and each mimic described by its similarity value.

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2.2 | Abundance-dependent predator recognition

In addition to the morphological similarity between the mimic and the model, we considered a novel predator recognition process that determines the predator's *realized* similarity $k_{\text{real},j}$ based on three factors. First, the realized similarity is constrained by the innate morphological similarity between mimics and the model, which corresponds to k_j in Equation 2 and represents the lowest value (highest distinguishability) that $k_{\text{real},j}$ can achieve. Second, the realized similarity is influenced by the abundance ratio of two prey categories: profitable ($\frac{v_i}{h_i} > 0$) and unprofitable ($\frac{v_i}{h_i} \le 0$) prey. Because $h_i > 0$ throughout, this classification is equivalently determined by the sign of v_i. When unprofitable prey are significantly more abundant than profitable prey, predators interacting with this mimicry complex are more likely to encounter unprofitable prey. This causes the predator to associate the shared morphological traits with the unprofitable prey, thereby perceiving both prey types as unprofitable prey. The opposite outcome occurs when the abundance of the profitable prey is significantly higher than that of the unprofitable prey—the predator will perceive both prey types as the more abundant profitable prey. However, when the two prey categories have a similar abundance, the predator will encounter an equal amount of both prey types; the resulting realized similarity will not be biased by prey abundance imbalance but instead be determined by the innate mimic–model morphological similarity. Finally, a third factor, σ_j, controls the sensitivity of the predator recognition to the ratio of prey abundance. To represent this predator recognition process, we formulated the following abundance-dependent recognition function, which phenomenologically characterizes how the realized similarity ($k_{\text{real},j}$) is affected by the three aforementioned factors:

$$k_{\text{real},j} = 1 - (1 - k_j) \times \exp\left[- \frac{\left(\log \frac{N_m + \sum_j (1-\delta_j) N_{c_j}}{\sum_j \delta_j N_{c_j}} \right)^2}{2\sigma_j^2} \right], \tag{3}$$

with $\delta_j = 0$ if N_{c_j} is unprofitable ($\frac{v_i}{h_i} \le 0$) and $\delta_j = 1$ if N_{c_j} is profitable ($\frac{v_i}{h_i} > 0$); the term within the logarithm thereby represents the abundance ratio of the two prey categories. In a two-mimicry system that contains a Batesian mimic (N_{c_1}) and a Müllerian mimic (N_{c_2}), the abundance ratio term becomes $(N_m + N_{c_2}) / N_{c_1}$. When the innate morphological similarity does not constrain realized similarity, for example easily distinguishable mimic with $k_j = 0.1$, $k_{\text{real},j}$ is greatly influenced by the abundance ratio of different prey categories and can vary freely between k_j and 1 (lower curve in Figure 1c). As the innate morphological similarity k_j approaches 1 (i.e. mimics become hardly distinguishable), the realized similarity $k_{\text{real},j}$ becomes less responsive to prey abundance ratio (upper curve in Figure 1c). The functional form of Equation (3) implies that $k_{\text{real},j}$ approaches 1 when the abundance of the unprofitable and profitable prey is highly unbalanced (i.e. when $\left(N_m + \sum_j (1-\delta_j) N_{c_j} \right) / \sum_j \delta_j N_{c_j}$ approaches 0 or ∞). Under such a scenario, the predator treats the two prey categories as if both were the more abundant prey category: with optimal foraging, both are treated as the profitable prey if the abundance ratio approaches 0, and both are treated as the unprofitable prey if the ratio approaches ∞.

Here, we interpret Equation (3) as a phenomenological representation of predator learning. Specifically, we assume that predators repeatedly sample prey, update their responses through associative learning and generalize those learned responses across prey with similar signals. As a result, when one prey category is much more abundant than the other, repeated encounters with that category can dominate predator experience and bias recognition towards the more common signal-sharing category, causing the realized recognition $k_{\text{real},j}$ approach 1. Under this interpretation, a rare prey item may be lumped into the dominant category rather than treated as entirely novel, which is plausible especially when the prey belong to the same general prey type and when learning costs are high (Kikuchi & Sherratt, 2015). We further assume that this abundance-dependent recognition operates only within the mimicry complex, whereas the alternative prey remains a distinct prey category by design. In other words, we envision a scenario in which the mimic and the model, although morphologically discriminable (i.e. $k \approx 0$), are still perceived as belonging to a broader shared category that is completely distinct from the alternative prey (see also Kikuchi & Sherratt, 2015; Sherratt, 2011).

2.3 | Mimicry dynamic system

Finally, we incorporated optimal foraging theory, signal detection theory and our novel abundance-dependent recognition function into a multi-mimicry dynamical system. We simulate the dynamics between mimics (N_{c_j}; $j = 1, 2, \cdots, n$), the model (N_m), the predator (N_p) and an alternative prey (N_n):

$$\frac{dN_p}{dt} = N_p \left[\frac{s\left(\sum_j N_{c_j} p_{c_j} v_{c_j} + N_m p_m v_m + N_n p_n v_n \right)}{1 + s\left(\sum_j N_{c_j} p_{c_j} h_{c_j} + N_m p_m h_m + N_n p_n h_n \right)} - \mu_p \right] \tag{4-1}$$

$$\frac{dN_{c_j}}{dt} = r_{c_j} N_{c_j} \left(1 - \frac{N_{c_j}}{K_{c_j}} \right) - N_p \left[\frac{s N_{c_j} p_{c_j}}{1 + s\left(\sum_j N_{c_j} p_{c_j} h_{c_j} + N_m p_m h_m + N_n p_n h_n \right)} \right] \tag{4-2}$$

$$\frac{dN_m}{dt} = r_m N_m \left(1 - \frac{N_m}{K_m} \right) - N_p \left[\frac{s N_m p_m}{1 + s\left(\sum_j N_{c_j} p_{c_j} h_{c_j} + N_m p_m h_m + N_n p_n h_n \right)} \right] \tag{4-3}$$

$$\frac{dN_n}{dt} = D(S_n - N_n) - N_p \left[\frac{s N_n p_n}{1 + s\left(\sum_j N_{c_j} p_{c_j} h_{c_j} + N_m p_m h_m + N_n p_n h_n \right)} \right]. \tag{4-4}$$

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The per capita population growth rate of the predator consists of the consumption gain from optimal foraging (first term within the bracket in Equation 4-1) and the density-independent mortality rate μ_p (second term within the bracket in Equation 4-1). Specifically, the first term captures the total foraging gain the predator obtains from consuming various prey types, where the optimal combination of p_i is instantaneously adjusted at each time step to maximize the consumption gain (Equation 1) under the constraints imposed by signal detection theory (Equation 2) and abundance-dependent recognition (Equation 3). The mimics and the model grow logistically, with intrinsic growth rates r_{c_j} and r_m, and carrying capacity K_{c_j} and K_m, respectively. The population of the mimics and the model decreases due to predator consumption, which depends on the optimized probability p_{c_j} and p_m, respectively (the term within brackets represents per predator consumption). Finally, we assumed that the alternative prey follows chemostat dynamics with a continuous flux (D) and an external supply source (S_n). Although the alternative prey is still consumed by the predator with optimized probability p_n, it represents a background prey pool outside the focal mimicry complex and is continuously replenished.

2.4 | Numerical simulations

We simulated two different mimicry systems: (1) a single-mimic system with only the Batesian mimic (N_c), and (2) a multi-mimic system with both Batesian mimic (N_{c_1}) and Müllerian mimic (N_{c_2}). For simplicity, we omitted the secondary subscript j in N_{c_j} in the single-mimic model, but reintroduced it in the multi-mimic system (see Appendix A for the full equations). We used the fourth-order Runge–Kutta method from `R` package `deSolve` (Soetaert et al., 2010) to numerically solve our dynamical system. In all simulations, the integration step size was set to 0.02 and the dynamics were simulated for 12,000 time steps. To incorporate optimal foraging, the values of p_i were adjusted at each integration step by finding the optimal combination that maximizes foraging gain (i.e. Equation 1). To better capture the nonlinear function form of Equation 2, the maximum foraging gain was found by evaluating Equation 1 with a range of p_m (and corresponding p_{c_j} based on each mimic's similarity k_j) values from 0 to 1 in increments of 0.0001. While a larger increment for p_m (e.g. 0.1) could accelerate the numerical procedure, we note that it would lead to insufficient sampling along the nonlinear curve. Parameters in the single-mimic system are as follows (note again secondary subscripts j were omitted): $v_c = 1.6, v_n = 0.8, v_m = 0, h_c = h_n = h_m = 1$. While prey defence can in principle affect both h_i and v_i, here we assumed that the model differs from other prey items primarily in v_i. The intrinsic growth rate and carrying capacity of the mimic and the model are $r_c = r_m = 2$ and $K_c = K_m = 10$, respectively, and the death rate of the predator is $\mu_p = 0.75$. The influx of the alternative prey is controlled by parameters $S_n = 20$ and $D = 1$, and the search rate is $s = 1$. We used the following initial conditions for all simulations: $N_p(0) = 0.5, N_c(0) = 1, N_m(0) = 10, N_n(0) = 20$. These initial conditions represent a mimicry complex embedded in a larger background pool of alternative prey. The parameters for the multi-mimics system are set as follows (secondary subscripts j were reintroduced to distinguish the two mimics): $r_{c_1} = r_{c_2} = r_m = 2$; $K_{c_1} = K_{c_2} = K_m = 10$; $v_{c_1} = 2.5, v_{c_2} = v_m = 0, v_n = 0.8$; $h_{c_1} = h_{c_2} = h_m = h_n = 1$; $\mu_p = 0.75$ and $S_n = 20$. With $v_{c_2} = 0$, N_{c_2} is the Müllerian mimic in the system, while N_{c_1} is the Batesian mimic with the highest profitability. We chose v_{c_1} so that the Batesian mimic remained the most profitable prey item, while the combined profitability of the fully lumped mimicry complex remained comparable to that of the alternative prey. The initial conditions for the multi-mimics system are identical to those of the single-mimicry system, with $N_{c_1}(0) = N_{c_2}(0) = 1$.

To study the effect of mimic–model morphological similarity on community dynamics in the single-mimic system, we explored k across the parameter range of 0 to 1 by an interval of 0.01. For both the single-mimic and multi-mimicry systems, we studied the effect of abundance-dependent recognition (Equation 3) by varying the morphological similarity (i.e. k for the single-mimic system and both k_1 and k_2 for the multi-mimic system) from 0 to 1. All simulations were carried out using R 4.3.3 (R Core Team, 2024). When presenting simulation results, the last 80% of each simulated time series was used to calculate the mean and variance of each species' abundance, which were then used to determine the stability and coexistence outcome of the system. Specifically, we set the extinction threshold for each species at 10^{-10}. The stability threshold was set at 10^{-12}; that is, the system was considered stable if the variance of the predator population is less than 10^{-12}. Since the predator interacts with all other prey and exhibits the highest population abundance, calculating the variance of the predator population is sufficient to determine the stability of the system.

3 | RESULTS

3.1 | Single Batesian mimic system

3.1.1 | Predator decision depending only on mimic–model morphological similarity

We first explored the dynamics of a single Batesian mimicry system where the predator's decision depends only on the morphological similarity between the mimic and the model (k; Figure 2a). We showed that along the spectrum of mimic–model similarity (k ranging from 0 to 1), low and high similarity produced cyclic dynamics while stable population dynamics were observed under intermediate similarity. Moreover, intermediate similarity ($0.5 < k < 0.8$) also resulted in higher predator and mimic abundance (red and orange, respectively, in Figure 2a upper panel), suggesting that intermediate similarity benefits the mimicry system in terms of stability and species abundance. Mechanistically, this increase arises because higher similarity makes attacks on the mimic increasingly costly, as such attacks are more likely to be accompanied by accidental attacks on the unprofitable model. As a result, optimal foraging reduces the predator's attack probability on the mimic and shifts foraging towards the

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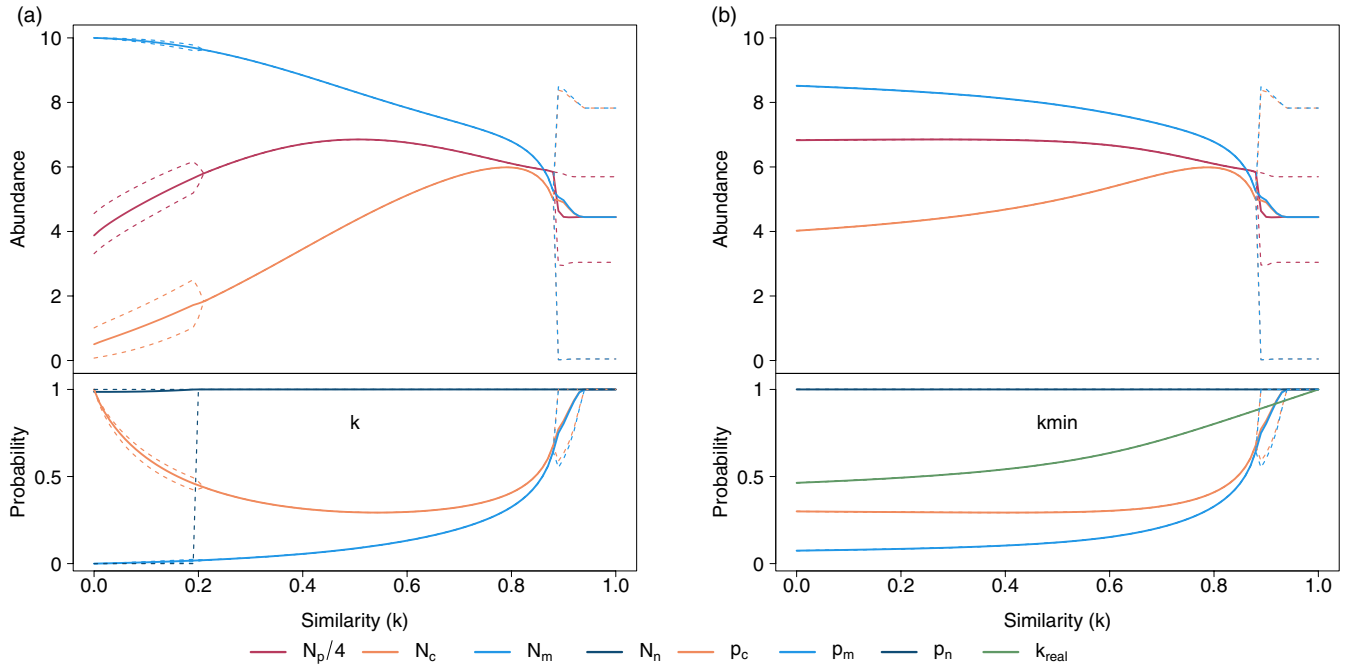


FIGURE 2 Bifurcation plot for the single-mimic system, with predator recognition process based on (a) mimic–model morphological similarity or (b) abundance-dependent recognition. Both (a) and (b) bifurcate along the innate mimic–model morphological similarity (k). The upper panels of both plots illustrate the abundance of the predator (N_p; red), the mimic (N_c; orange) and the model (N_m; light blue). Note that the abundance of the predator is divided by four for better visualization. The solid line represents the mean abundance, and the dashed line represents the maximum and minimum of the abundance. The lower panels of both plots illustrate the predator's attack probability on the mimic (p_c; orange), the model (p_m; light blue) and the alternative prey (p_n; dark blue). In panel (b), the realized recognition between the mimic and the model (k_{real}; Equation 3) is plotted in green. The solid and dashed lines in the lower panels also represent the mean and maximum/minimum values, respectively. Note that the lines overlap at high values of k; a version with alternative prey abundance is shown in Figure S2.

alternative prey, thereby stabilizing the system at higher predator and mimic abundance under intermediate similarity. We discuss the dynamics below (see also Figure 3 for detailed time series under different k values).

When the similarity between the mimic and the model is zero (i.e. $k = 0$, left-most value in Figure 2a), the mimic and the model are treated as two different species by the predator, and the system reduces to a classical optimal foraging scenario among three distinct prey types. In this scenario, cycles emerge from diet-composition shifts: while predators always ignore the unprofitable model ($\frac{v_m}{h_m} = 0$) and always attack the mimic due to its high profitability, predators switch between, including or excluding the alternative prey from its diet since changes in mimic abundance lead to different optimal p_n (Figure 3a, where thick blue bands reflect rapid switching between $p_n = 0$ and $p_n = 1$ during diet-composition shifts; see also Appendix B for analytical derivation and Figures S1 and S2). With slightly increased mimic–model similarity (i.e. increasing k towards approximately 0.2), the probability of the model being attacked increased while the attack probability of the mimic decreased (light blue and orange, respectively, in the lower panel of Figure 2a). This is because predators now sometimes cannot differentiate between the mimic and the model and consume the wrong prey. Under such low-similarity scenario ($0 < k < 0.2$), we observed that the system still exhibits cycles in its diet composition. However, unlike the scenario when $k = 0$, where predators exclude the model from their diet, here all three prey are included (Figure 3b), demonstrating that even a slight similarity between the mimic and model can influence the community dynamics.

As the mimic–model similarity continues to increase (as k approaches 0.5), predators find it increasingly difficult to distinguish between the mimic and the model. Under this scenario, maintaining a high attack probability towards the mimic becomes increasingly costly because attacks on the mimic are more likely to be accompanied by accidental consumption of the unprofitable model. Optimal foraging therefore leads predators to reduce their attack probability on the mimic. This foraging decision reduces the cost of mistakenly consuming the model and leads to more alternative prey consumption, ultimately stabilizing the predator population at higher abundance (Figure 3c; Figures S2 and S3). At the same time, the mimic population increases in abundance due to the released predation pressure, while the model population continues to decline because it is still attacked through its linkage to the mimic. Surprisingly, as the similarity between the mimic and the model increases further ($0.6 < k < 0.8$), the probability of attacking the mimic rises again, producing a U-shaped relationship between mimic–model similarity and attack probability across this intermediate range of k. This reversal occurs because while predators could theoretically avoid accidentally consuming the model by further reducing attacks on mimics,

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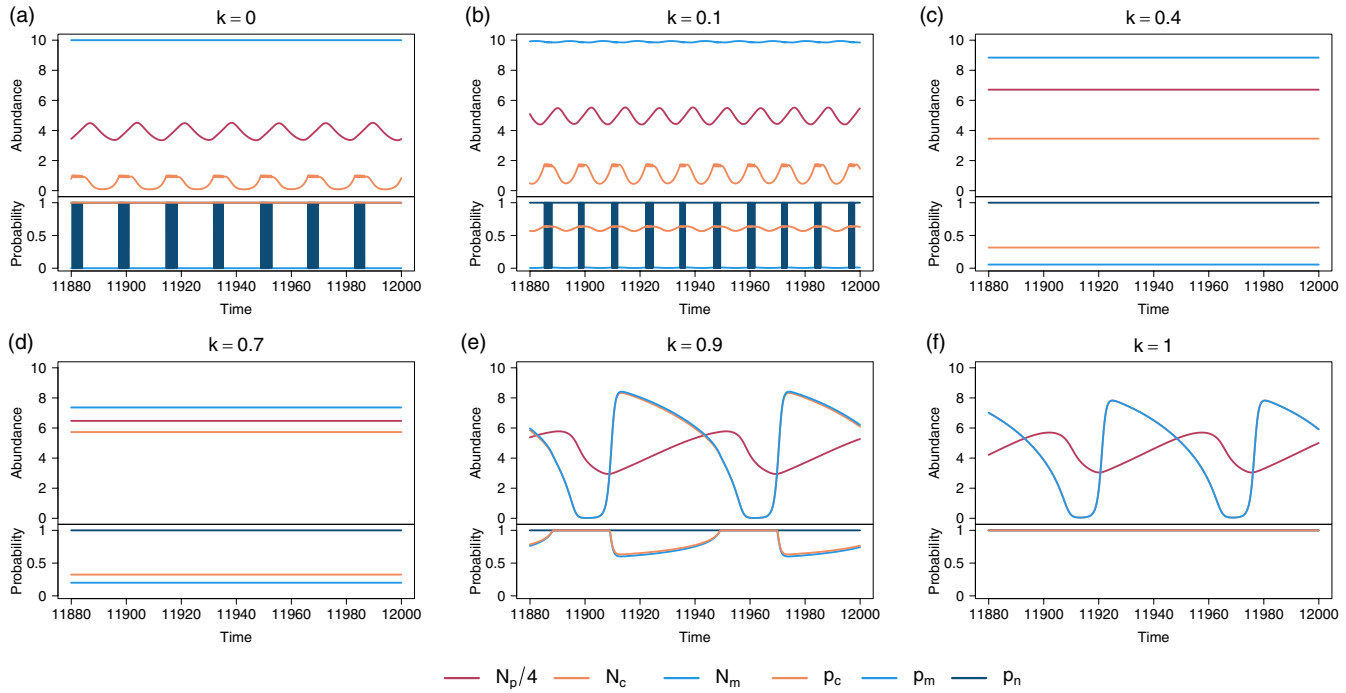


FIGURE 3 Time series of species abundance and predator attack probability for the single-mimic system with predator recognition based solely on mimic–model morphological similarity (k). Similarity (k) values vary as follows: (a) $k = 0$, (b) $k = 0.1$, (c) $k = 0.4$, (d) $k = 0.7$, (e) $k = 0.9$ and (f) $k = 1$. The upper half of each panel illustrates the abundance of the predator (N_p; red), the mimic (N_c; orange) and the model (N_m; light blue). Note that the abundance of the predator is divided by four for better visualization. The lower half of each panel illustrates the predator's probability of attacking the mimic (p_{c1}; orange), the model (p_m; light blue) and the alternative prey (p_n; dark blue). Note in panel (f) the lines representing the mimic and the model completely overlap as they are perceived as the same species under $k = 1$.

optimal foraging prevents this strategy from being realized as the predator still requires foraging gain from the mimic–model species pair. Instead, predators unavoidably need to increase their attack probability on both prey items to make up for the frequent unprofitable accidental attacks, thereby decreasing the abundance of the predator (Figure S3). Initially, this decline in predator abundance allows the mimic population to rise further, reaching its peak just before similarity exceeds $k = 0.8$ (Figure 3d). However, once similarity surpasses this threshold, the rising attack probability on mimics leads to a decrease in mimic abundance. Consequently, the peaks in predator and mimic abundance occur at different values of k, reflecting the shifting balance between prey profitability and foraging pressure within the mimicry complex.

In addition to the counterintuitive increase in p_c, a higher mimic–model similarity also results in the reoccurrence of cyclic dynamics ($0.88 < k < 1$; right-most region in Figure 2a). When $0.88 < k < 0.94$, population cycles are characterized by diet-composition shifts (Figure 3e) similar to those seen under low mimic–model similarity. However, once similarities surpass a certain threshold ($k > 0.93$), the cycles are akin to classic predator–prey population cycles and no longer involve diet-composition shifts. Instead, predators now view the mimic–model species pair as a single species and consistently include both in their diet ($p_m = p_c = 1$; Figure 3f). Note that despite their inability to distinguish the mimic and the model, predators do not discard the mimic–model species pair from their diet because its collective value remains profitable. We show in Figure S4 that making $v_m < 0$ can, heuristically, reduce the collective profitability of the mimic–model species pair, eventually causing predators to discard the species pair altogether if they share high morphological similarity (Figure S4).

3.1.2 | Predator decision depending on abundance-dependent recognition

We next relaxed the assumption that the predator's decision process is solely determined by mimic–model morphological similarity (k), making it dependent on the abundance ratio of two prey categories, namely, the profitable mimic and the unprofitable model ($\frac{N_m}{N_c}$; Equation 3). We used the notation k_{real} to represent the realized similarity between the mimic and the model from the perspective of the predator (again, secondary subscript omitted for simplicity). Unlike the previous scenario (Figure 2a), which produced diet shift cycles under low similarity values, abundance-dependent recognition leads to stable dynamics under a wide range of innate similarity values ($0 < k < 0.88$; Figure 2b). This suggests that when the innate mimic–model similarity is not too high, the predator recognition is influenced by the observed $\frac{N_m}{N_c}$ ratio and reaches an intermediate realized similarity (green line in Figure 2b lower panel). The realized intermediate similarity ($0.46 < k_{\text{real}} < 0.88$) corresponds to values in Figure 2a that stabilize the system and lead to higher predator and mimic abundance. In other words, abundance-dependent recognition

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leads predators to reduce attack on mimics with low morphological similarity, causing the system to exhibit stable dynamics associated with intermediate similarity values (i.e. $k_{real} > k$).

However, the predator becomes less responsive to the prey abundance ratio, and the realized similarity becomes increasingly similar to k, as predator recognition becomes increasingly constrained by the innate mimic–model similarity (i.e. high k). That is, the realized similarity cannot reach an intermediate value that would stabilize the system. Instead, the high innate mimic–model morphological similarity causes the predator to perceive the two prey as a single species, which causes $k_{real} \approx k$ as $\frac{N_m}{N_c} \approx 1$; the resulting dynamics is a cyclic behaviour with reduced population sizes (corresponding to a high k scenario in Figure 2a).

3.2 | Multi-mimic system with abundance-dependent recognition

We expanded our model to consider a multi-mimic mimicry system consisting of a Batesian mimic (N_{c_1}), a Müllerian mimic (N_{c_2}) and a model species (N_m), with the latter two considered as unprofitable prey ($v_{c_2} = v_m = 0$). We considered this community composition as it represents the simplest mimicry complex with multiple types of mimicries (see also Figure S5 for an example with more species). We explored how the Batesian mimic–model morphological similarity (k_1) and the Müllerian mimic–model morphological similarity (k_2) influenced community dynamics, assuming the predator possesses abundance-dependent recognition (see also Figure S6 for the case where foraging decisions depend solely on mimic–model morphological similarity). Intuitively, a high $k_{real,1}$ value causes the predator to perceive the mimicry complex as a more profitable group of prey, whereas a high $k_{real,2}$ would decrease the overall profitability of the mimicry complex. Briefly, the simulation reveals that while intermediate similarity in both Batesian and Müllerian mimicry stabilizes the system and increases the abundance of the predator and mimics, different types of mimicry vary in their impact on the species abundance (Figure 4).

Our simulations suggest that under abundance-dependent recognition, the system exhibits two distinct dynamical regimes, governed by the interplay between k_1 and k_2. The top-left regime of Figure 4 is characterized by lower attack probability towards the mimicry complex (i.e. lower p_{c_1} and p_{c_2}; top-left of Figure 4b,d) and higher abundance of N_{c_1}, N_{c_2} and N_p (top-left of Figure 4a,c,g). In contrast, in the bottom-right regime of Figure 4, the Batesian and the Müllerian mimics are consistently included in the diet with high attack probability, leading to their lower abundance (lower-right of Figure 4a–d). Additionally, under abundance-dependent recognition, most parameter combinations in the top-left regime exhibit stable population dynamics, while parameter combinations in the lower-right regime exhibit cyclic dynamics (Figure 4h). The boundary separating the two regimes shows a positive association between k_1 and k_2, indicating that a higher value of k_2 allows the system to tolerate a greater k_1 before becoming destabilized (Figure 4h). When k_1 is sufficiently high, the system eventually becomes unstable as the predator increasingly perceives the mimicry complex as a profitable food source, akin to the instability seen in the single Batesian mimic system with high similarity (Figure 2b). However, the presence of a Müllerian mimic can mitigate this effect: a Müllerian mimic with sufficiently high k_2 can stabilize the system, especially when k_1 is fixed at low to intermediate values (Figure 4h). This demonstrates that mimic types differ in their impact on system stability, with a perfect Batesian mimic (high k_1) causing the system to be unstable and a perfect Müllerian mimic (high k_2) promoting stability.

Beyond stability, we examined how the Batesian mimic–model morphological similarity (k_1) affects mimic abundance while keeping the Müllerian mimic's morphological similarity (k_2) constant. Under abundance-dependent recognition, $k_{real,1}$ remains within an intermediate range ($0.5 < k_{real,1} < 0.8$; Figure 4e), and Batesian mimic abundance exhibits a hump-shaped pattern with increasing k_1 (orange line in Figure 5a). Moreover, compared with the scenario where predator recognition relies only on morphological similarity (Figure S6a), the resulting intermediate $k_{real,1}$ leads to a higher Batesian mimic abundance. The Müllerian mimic abundance, on the other hand, decreases monotonically with increasing k_1 (green line in Figure 5a). As k_1 further increases, the dynamics eventually transition into the aforementioned bottom-right regime, characterized by unstable dynamics with consistently low abundance of both mimics as the predator now consumes the entire mimicry complex as a single palatable prey.

Finally, we examined how varying the Müllerian mimic–model morphological similarity (k_2) affects mimic abundance, while keeping the Batesian mimic's morphological similarity (k_1) fixed. When the Batesian mimic has an innate intermediate similarity ($0.5 < k_1 < 0.8$), the system is likely within the regime with low abundance of both mimics and cyclic dynamics (Figure 4a,c). Under this scenario, a Müllerian mimic with high similarity provides protection to the mimicry complex: higher k_2 balances out the impact from a highly similar Batesian mimic, resulting in the transition into the regime with high mimic abundances and stable dynamics. Moreover, within the top-left regime, while $k_{real,1}$ settles within an intermediate level ($0.5 < k_{real,1} < 0.8$) and the Batesian mimic thereby maintains higher abundance, $k_{real,2}$ increases monotonically with increasing k_2 (Figure 4f). As a result, Müllerian mimic abundance increases with k_2 (Figure 5b), suggesting that Müllerian mimicry provides more protection under higher morphological similarity. Together, these results underscore the contrasting ecological roles of the two mimic types. For the Batesian mimic, intermediate similarity to the model reduces predation while avoiding the high k_1 that causes predators to consume the entire mimicry complex as a single profitable prey. In contrast, Müllerian mimics benefit from high similarity as it reinforces predator learning, thereby leading to a steady increase in abundance with increasing k_2.

4 | DISCUSSION

We showed that imperfect mimicry (i.e. intermediate similarity between the mimic and the model) promotes stable ecological dynamics and allows the mimic to reach high abundance in a single Batesian

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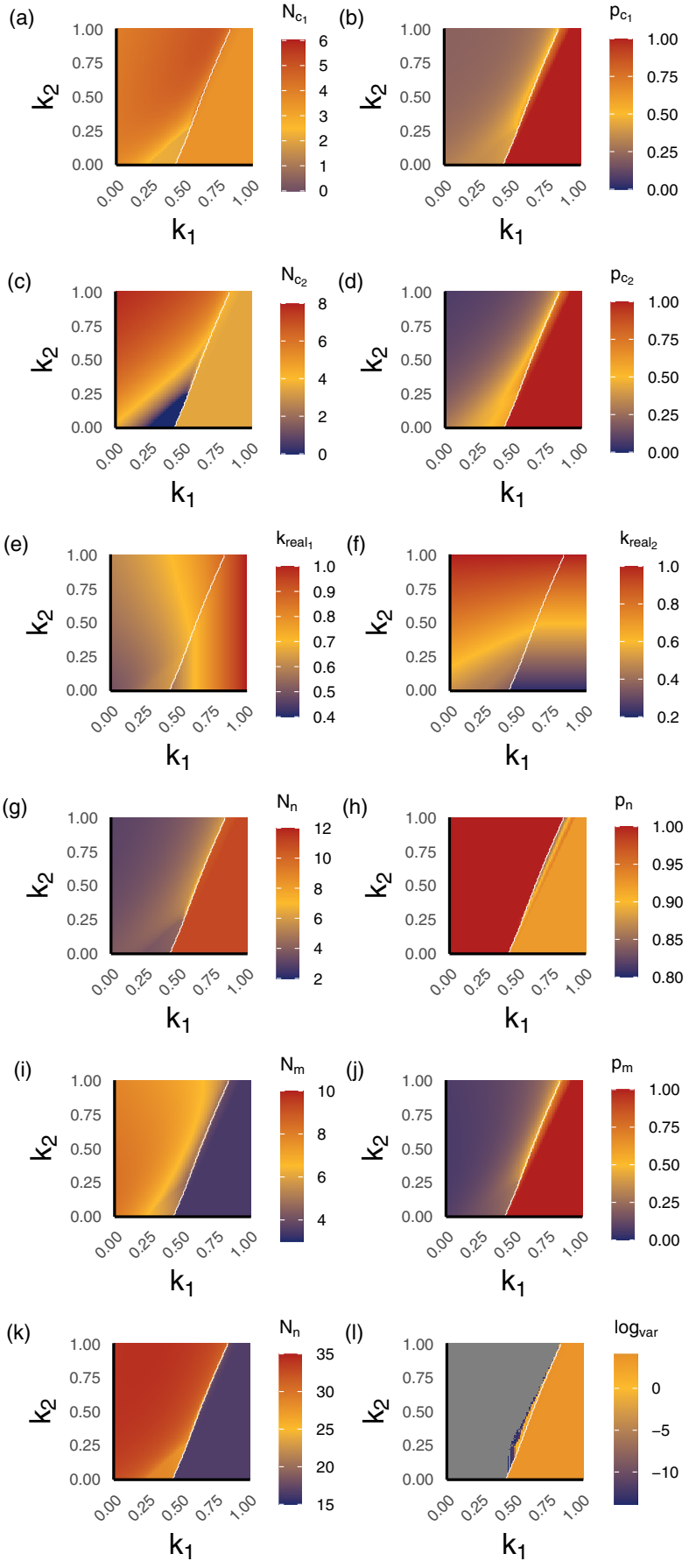


FIGURE 4 Effect of the Batesian mimic–model morphological similarity (k_1; *x*-axis) and the Müllerian mimic–model morphological similarity (k_2; *y*-axis) on system dynamics and species abundances. The different panels represent different variables: (a) Batesian mimic abundance (N_{c_1}), (b) attack probability on the Batesian mimic (p_{c_1}), (c) the Müllerian mimic abundance (N_{c_2}), (d) attack probability on the Müllerian mimic (p_{c_2}), (e) realized Batesian mimic similarity ($k_{real,1}$), (f) realized Müllerian mimic similarity ($k_{real,2}$), (g) alternative prey abundance (N_n), (h) attack probability on the alternative prey (p_n), (i) model abundance (N_m), (j) attack probability on the model (p_m), (k) predator abundance (N_p) and (l) the log(variance) of the predator population fluctuation, serving as an indicator of community stability. The white outline in all panels depicts the proposed boundary separating the two dynamical regimes (see Appendix B). Note that the abundances of different state variables are on different scales to better show the pattern of each variable.

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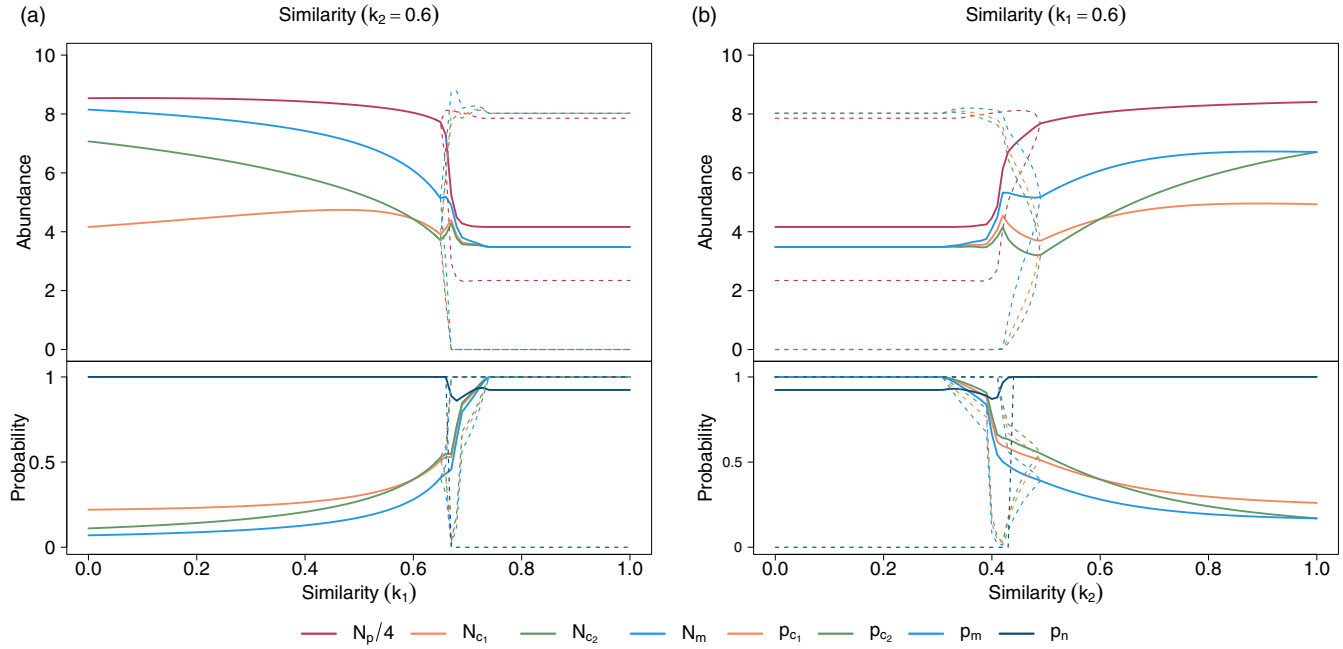


FIGURE 5 Bifurcation plot of the multi-mimicry system. (a) bifurcation along the Batesian mimic–model morphological similarity (k_1) while fixing the Müllerian mimic's morphological similarity ($k_2 = 0.6$). (b) bifurcation along the k_2 while fixing $k_1 = 0.6$. The *y*-axis of the upper panels shows abundance, and the lower panels show the attack probability. The upper panels of both plots illustrate the abundances of the predator (N_p; red), the Batesian mimic (N_{c_1}; orange), the Müllerian mimic (N_{c_2}; green) and the model (N_m; light blue). Note that the abundance of the predator is divided by four for better visualization. The lower panels of both plots illustrate the predator's attack probability on the Batesian mimic (p_{c_1}; orange), the Müllerian mimic (p_{c_2}; green), the model (p_m; light blue) and the alternative prey (p_n; dark blue). The solid and dashed lines in all panels represent the mean and maximum/minimum values, respectively.

mimicry system. This outcome holds whether predator recognition is shaped solely by morphological similarity or by a combination of morphological similarity and prey abundance ratio (Figure 2). In a multi-mimicry system, including both Batesian and Müllerian mimics, intermediate similarity remains beneficial for the Batesian mimic, while high similarity for the Müllerian mimic leads to increased mimic abundance (Figure 4). These contrasting patterns suggest that the two mimic types have different impacts on the system: the Batesian mimic can destabilize the system (and lead to low mimic abundance) when too similar to the model, whereas the Müllerian mimic with high similarity can promote system stability (and lead to high mimic abundance). Together, our results suggest that imperfect mimicry can lead to ecologically stable dynamics across different mimicry systems, offering new insights into how mimicry persists in nature.

Our results provide a mechanistic perspective on the 'eye of the beholder hypothesis' (Cuthill & Bennett, 1993; Dittrich et al., 1993), which proposes that mimicry deemed imperfect by human observers may be functionally effective from the perspective of the predator. Specifically, our framework assumes a biologically motivated abundance-dependent recognition process in which repeated encounters with an abundant prey category bias predators to generalize their responses across signal-sharing prey. Under this assumed mechanism, predators may exhibit a higher level of realized similarity than would be expected from morphology alone (i.e. $k_{\text{real},j} \geq k_j$). This formulation echoes Müller's original idea that the inclusion of more unprofitable prey strengthens the protective benefits of aposematic signals (Müller, 1879), and it aligns with the principles of Pavlovian conditioning, in which repeated signal exposures reinforce learned associations. Computer-based foraging experiments with human participants have shown that prey community structure can shift the prey categorization behaviour of the predator (Kikuchi et al., 2019; McGuire et al., 2006), supporting the broader plausibility that predator cognition can depend on ecological context. By incorporating prey abundance into predator recognition and mimicry dynamics, our framework extends beyond static morphological similarity and highlights an ecological dimension that human observers typically overlook. This mismatch between predator and human perception may help explain how the 'eye of the beholder' effect could emerge from predator decision-making behaviour.

We note that our abundance-dependent recognition function should be interpreted as one plausible phenomenological form of context-dependent recognition. Although predator sensitivity to this process can vary through the parameter σ_j (Figure S7), other forms of predator recognition not captured by Equation 3 are also biologically possible. For example, rare mimics with low similarity might be sampled separately rather than lumped into the more common category (Kikuchi & Sherratt, 2015; Sherratt, 2011), or avoided altogether because of dietary conservatism (Aubier & Sherratt, 2015); both possibilities differ from the assumption in our model. Experimental studies also suggest that ecological context can alter prey categorization in ways that differ from the pattern assumed here. For example, Kikuchi et al. (2019) showed that greater community diversity

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can promote broader prey categorization rather than better discrimination. Exploring such alternative formulations would be a valuable direction for future theory and could help clarify the role of abundance-dependent recognition in mimicry systems. This issue may be particularly important when the mimic–model pair has low similarity, where context-dependent recognition could influence whether initially weak resemblance is ecologically sufficient for mimic populations to persist.

Our results from the single Batesian mimicry system illustrate that imperfect mimicry could promote mimic abundance over ecological timescales, suggesting that intermediate resemblance may be ecologically advantageous in simple Batesian systems. This ecological pattern is broadly consistent with recent empirical work showing that imperfect mimicry is more evolutionarily stable than perfect mimicry (Kelly et al., 2025), although an explicit eco-evolutionary model would be needed to connect these abundance outcomes directly to selection on mimic similarity. In contrast, our results from the multi-mimicry system suggest that the ecological consequences of mimic similarity depend on the composition of the mimicry complex. These results raise the possibility that the composition of a mimicry complex influences which levels of mimic–model similarity can be maintained under a given set of ecological conditions. Empirical studies also indirectly support this idea: Müllerian mimicry rings could include dozens of species with varying degrees of resemblance and even overlapping traits across different mimicry rings (Motyka et al., 2020; Wilson et al., 2012, 2022). Future work can incorporate evolutionary dynamics into our framework to study how the presence of Batesian mimics influences the range of warning signals maintained in Müllerian mimicry rings.

The ecological patterns identified by our model could be tested by comparative analysis of trait data and mimicry-complex composition. One can analyse morphological traits, such as colour, pattern or other aposematic signals to characterize the similarity between mimics and models (Eliason et al., 2019; Kelly et al., 2021; Maia et al., 2019), thereby providing a basis to evaluate whether empirical systems align with theoretical predictions. For example, a meta-analysis of mimic–model morphological similarity could examine whether single Batesian mimicry systems tend to exhibit intermediate similarity to the model. Moreover, these ecological patterns could be tested through behavioural experiments that assess predator attack probabilities across different mimic species with varying degrees of similarity to their models (Tseng et al., 2014). Such experiments could also test whether predator responses to mimic similarity depend on the recognition process or ecological setting. More specifically, experiments that manipulate the relative abundance or encounter frequency of prey types could be used to test the abundance-dependent recognition mechanism directly.

Finally, we encourage future work to explicitly integrate aposematic traits into the theoretical framework of mimicry dynamics (Holen & Johnstone, 2004; Tomizuka & Tachiki, 2024). One promising approach is to replace the similarity parameter in our model with species trait distance in a multidimensional trait space. For example, rather than projecting trait differences onto a single model-centred similarity axis, one could link traits more directly to predator attack probabilities. Such a trait-explicit framework would provide a more mechanistic interpretation of mimic–model similarity and avoid potential artefacts arising from the phenomenological linkage between attack probabilities in signal detection theory. For instance, in our current framework, the attack probabilities of the two mimics are indirectly constrained by their linkage to the model's attack probability (see Appendix C), which can create abrupt abundance drops at low similarity values that are unlikely to reflect a biologically realistic outcome of predator recognition (Figure 4c). A trait-explicit framework would also allow one to incorporate pairwise resemblance among mimics, enabling the investigation of more complex similarity relationships that better represent natural mimicry systems. Furthermore, it would allow the investigation of eco-evolutionary feedback dynamics (Tomizuka & Tachiki, 2024), where mimic trait distributions evolve under selection imposed by predators. Such a framework could unify the ecological and evolutionary processes shaping imperfect mimicry, thereby clarifying not only its ecological stability but also its evolutionary persistence. Overall, our study underscores the significance of considering predator recognition and population dynamics into mimicry theory, offering new insights into how imperfect mimicry can be maintained in natural, complex mimicry systems.

AUTHOR CONTRIBUTIONS

Y.S. conceived the study and conducted the simulations with assistance from P.-J.K., Y.S. and P.-J.K. wrote the manuscript. Both authors contributed to the final version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest.

DATA AVAILABILITY STATEMENT

All simulation R-scripts used to generate the results are archived on Zenodo with a permanent: https://doi.org/10.5281/zenodo.17234680.

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REFERENCES

Akcali, C. K., Kikuchi, D. W., & Pfennig, D. W. (2018). Coevolutionary arms races in Batesian mimicry? A test of the chase-away hypothesis. *Biological Journal of the Linnean Society*, *124*, 668–676.

Aubier, T., Joron, M., & Sherratt, T. (2017). Mimicry among unequally defended prey should be mutualistic when predators sample optimally. *The American Naturalist*, *189*, 267–282.

Aubier, T. G., & Sherratt, T. N. (2015). Diversity in müllerian mimicry: The optimal predator sampling strategy explains both local and regional polymorphism in prey. *Evolution*, *69*, 2831–2845.

Bates, H. W. (1862). Contributions to an insect fauna of the amazon valley. Lepidoptera: Heliconidae. *Transactions of the Linnean Society of London*, *23*, 495–566.

Bosque, R. J., Lawrence, J. P., Buchholz, R., Colli, G. R., Heppard, J., & Noonan, B. (2018). Diversity of warning signal and social interaction influences the evolution of imperfect mimicry. *Ecology and Evolution*, *8*, 7490–7499.

Brower, L. P., & Moffitt, C. M. (1974). Palatability dynamics of cardenolides in the monarch butterfly. *Nature*, *249*, 280–283.

Charnov, E. (1973). *Optimal foraging: Some theoretical explorations*. University of Washington.

Charnov, E. L. (1976). Optimal foraging: Attack strategy of a mantid. *The American Naturalist*, *110*, 141–151.

Chittka, L., & Osorio, D. (2007). Cognitive dimensions of predator responses to imperfect mimicry. *PLoS Biology*, *5*, 1–5.

Cuthill, I. C., & Bennett, A. T. D. (1993). Mimicry and the eye of the beholder. *Proceedings of the Royal Society B: Biological Sciences*, *253*, 203–204.

Darst, C. R. (2006). Predator learning, experimental psychology and novel predictions for mimicry dynamics. *Animal Behaviour*, *71*, 743–748.

Dittrich, W., Gilbert, F., Green, P., Mcgregor, P., & Grewcock, D. (1993). Imperfect mimicry: A pigeon's perspective. *Proceedings of the Royal Society B: Biological Sciences*, *251*, 195–200.

Edmunds, M. (2000). Why are there good and poor mimics? *Biological Journal of the Linnean Society*, *70*, 459–466.

Eliason, C. M., Andersen, M. J., & Hackett, S. J. (2019). Using historical biogeography models to study color pattern evolution. *Systematic Biology*, *68*, 755–766.

Franks, D. W., Ruxton, G. D., & Sherratt, T. N. (2009). Warning signals evolve to disengage batesian mimics. *Evolution*, *63*, 256–267.

Getty, T. (1985). Discriminability and the sigmoid functional response: How optimal foragers could stabilize model-mimic complexes. *The American Naturalist*, *125*, 239–256.

Holen, O., & Johnstone, R. (2004). The evolution of mimicry under constraints. *The American Naturalist*, *164*, 598–613.

Huheey, J. E. (1964). Studies of warning coloration and mimicry. IV. a. mathematical model of model-mimic frequencies. *Ecology*, *45*, 185–188.

Kelly, M. B., Derkarabetian, S., McLean, D. J., Shofner, R., Grismado, C. J., Haddad, C. R., Cassis, G., Giribet, G., Herberstein, M. E., & Wolff, J. O. (2025). Batesian mimicry converges toward inaccuracy in myrmecomorphic spiders. *Systematic Biology*, *74*, 967–984.

Kelly, M. B., McLean, D. J., Wild, Z. K., & Herberstein, M. E. (2021). Measuring mimicry: Methods for quantifying visual similarity. *Animal Behaviour*, *178*, 115–126.

Kikuchi, D. W., Barfield, M., Herberstein, M. E., Mappes, J., & Holt, R. D. (2022). The effect of predator population dynamics on batesian mimicry complexes. *The American Naturalist*, *199*, 406–419.

Kikuchi, D. W., Dornhaus, A., Gopeechund, V., & Sherratt, T. N. (2019). Signal categorization by foraging animals depends on ecological diversity. *eLife*, *8*, e43965.

Kikuchi, D. W., Pfennig, D. W., & Dykhuizen, H. E. D. (2013). Imperfect mimicry and the limits of natural selection. *Quarterly Review of Biology*, *88*, 297–315.

Kikuchi, D. W., & Sherratt, T. N. (2015). Costs of learning and the evolution of mimetic signals. *The American Naturalist*, *186*, 321–332.

Kumazawa, F., Asami, T., Hayashi, T., & Yoshimura, J. (2006). Population dynamics of batesian mimicry under interspecific competition. *Evolutionary Ecology Research*, *8*, 591–604.

Lindström, L., Alatalo, R. V., Lyytinen, A., & Mappes, J. (2004). The effect of alternative prey on the dynamics of imperfect Batesian and Müllerian mimicries. *Evolution*, *58*, 1294–1302.

Maia, R., Gruson, H., Endler, J. A., & White, T. E. (2019). Pavo 2: New tools for the spectral and spatial analysis of colour in R. *Methods in Ecology and Evolution*, *10*, 1097–1107.

Mallet, J., & Joron, M. (1999). Evolution of diversity in warning color and mimicry: Polymorphisms, shifting balance, and speciation. *Annual Review of Ecology and Systematics*, *30*, 201–233.

McGuire, L., Van Gossum, H., Beirinckx, K., & Sherratt, T. N. (2006). An empirical test of signal detection theory as it applies to Batesian mimicry. *Behavioural Processes*, *73*, 299–307.

McLean, D. J., Cassis, G., Kikuchi, D. W., Giribet, G., & Herberstein, M. E. (2019). Insincere flattery? Understanding the evolution of imperfect deceptive mimicry. *The Quarterly Review of Biology*, *94*, 395–415.

Motyka, M., Bocek, M., Kusy, D., & Bocak, L. (2020). Interactions in multipattern Müllerian communities support origins of new patterns, false structures, imperfect resemblance and mimetic sexual dimorphism. *Scientific Reports*, *10*, 11193.

Müller, F. (1879). Ituna and Thyridia: A remarkable case of mimicry in butterflies. *Transactions of the Entomological Society of London*, *1879*, 20–29.

Nelson, D., Crossland, M., & Shine, R. (2011). Foraging responses of predators to novel toxic prey: Effects of predator learning and relative prey abundance. *Oikos*, *120*, 152–158.

Nur, U. (1970). Evolutionary rates of models and mimics in Batesian mimicry. *The American Naturalist*, *104*, 477–486.

Oaten, A., Pearce, C., & Smyth, M. (1975). Batesian mimicry and signal detection theory. *Bulletin of Mathematical Biology*, *37*, 367–387.

Pekár, S., Jarab, M., Fromhage, L., & Herberstein, M. E. (2011). Is the evolution of inaccurate mimicry a result of selection by a suite of predators? A case study using myrmecomorphic spiders. *The American Naturalist*, *178*, 124–134.

Penney, H. D., Hassall, C., Skevington, J. H., Abbott, K. R., & Sherratt, T. N. (2012). A comparative analysis of the evolution of imperfect mimicry. *Nature*, *483*, 461–464.

Pfennig, D. W., & Kikuchi, D. W. (2012). Competition and the evolution of imperfect mimicry. *Current Zoology*, *58*, 608–619.

Quicke, D. L. (2017). *Mimicry, crypsis, masquerade and other adaptive resemblances*. John Wiley & Sons.

R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.

Rowell-Rahier, M., Pasteels, J., Alonso-Media, A., & Brower, L. (1995). Relative unpalatability of leaf beetles with either biosynthesized or sequestered chemical defence. *Animal Behaviour*, *49*, 709–714.

Ruxton, G. D., Allen, W. L., Sherratt, T. N., & Speed, M. P. (2018). *Avoiding attack: The evolutionary ecology of crypsis, aposematism, and mimicry*. Oxford University Press.

Ruxton, G. D., Sherratt, T. N., & Speed, M. P. (2004). *Avoiding attack: The evolutionary ecology of crypsis, warning signals and mimicry*. Oxford University Press.

Schultze, W. (1923). A monograph of the pachyrrhynchid group of the Brachyderinae, Curculionidae: Part III. The genera apocyrtidius heller and metapocyrtus heller. *The Philippine Journal of Science*, *6*, 609–673.

Sherratt, T. N. (2002). The evolution of imperfect mimicry. *Behavioral Ecology*, *13*, 821–826.

Sherratt, T. N. (2008). The evolution of Müllerian mimicry. *Naturwissenschaften*, *95*, 681–695.

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Sherratt, T. N. (2011). The optimal sampling strategy for unfamiliar prey. *Evolution*, *65*, 2014–2025.

Sherratt, T. N., & Peet-Paré, C. A. (2017). The perfection of mimicry: An information approach. *Philosophical Transactions of the Royal Society, B: Biological Sciences*, *372*, 20160340.

Soetaert, K., Petzoldt, T., & Setzer, R. W. (2010). Solving differential equations in R: Package deSolve. *Journal of Statistical Software*, *33*, 1–25.

Stephens, D. W., & Krebs, J. R. (1986). *Foraging theory* (Vol. *1*). Princeton University Press.

Tomizuka, H., & Tachiki, Y. (2024). The eco-evolutionary dynamics of Batesian mimicry. *Journal of Theoretical Biology*, *577*, 111683.

Tseng, H. Y., Lin, C. P., Hsu, J. Y., Pike, D. A., & Huang, W. S. (2014). The functional significance of aposematic signals: Geographic variation in the responses of widespread lizard predators to colourful invertebrate prey. *PLoS One*, *9*, 1–7.

Vane-wright, R. I. (1980). On the definition of mimicry. *Biological Journal of the Linnean Society*, *13*, 1–6.

Wilson, J. S., Pan, A. D., Alvarez, S. I., & Carril, O. M. (2022). Assessing Müllerian mimicry in north American bumble bees using human perception. *Scientific Reports*, *12*, 17604.

Wilson, J. S., Williams, K. A., Forister, M. L., von Dohlen, C. D., & Pitts, J. P. (2012). Repeated evolution in overlapping mimicry rings among north American velvet ants. *Nature Communications*, *3*, 1272.

Yamauchi, A. (1993). A population dynamic model of Batesian mimicry. *Researches on Population Ecology*, *35*, 295–315.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1: Time series plots showing the (a) diet switching criterion and (b) probability of consuming the alternative prey (p_n) in a single Batesian mimicry system with $k = 0.1$. Here, we show the diet switching behaviour from only consuming the mimicry species pair (G_{mc}) to including the alternative prey (G_{mcn}). The x-axis shows time step 1181 to 1185 (with integration step size 0.02). The black line in panel A is the energy gain of attacking only the mimicry complex and the red line shows the profitability of the alternative prey (constant $\frac{v_n}{h_n}$; Equation S6). The blue line in panel B is the probability of attacking the alternative prey (p_n). Grey strips indicate the time period when the switching criterion is met, thereby inducing a diet switch. The parameter values are identical to those in the main text, see section Numerical simulations for details.

Figure S2: Bifurcation plot for the single-mimic system, with predator recognition process based on (a) mimic–model morphological similarity or (b) abundance-dependent recognition. Both (a) and (b) bifurcate along the innate mimic–model morphological similarity (k). The upper panels of both plots illustrate the abundance of the predator (N_p; red), the mimic (N_c; orange), the model (N_m; light blue) and the alternative prey (N_n, dark blue). Note that the abundance of the predator is divided by four for better visualization. The dash-dotted line represents the mean abundance, and the dotted line represents the maximum and minimum of the abundance. The lower panels of both plots illustrate the predator's attack probability on the mimic (p_c; orange), the model (p_m; light blue) and the alternative prey (p_n; dark blue). In panel (b), the realized recognition between the mimic and the model (k_{real}; Equation 3) is plotted in green. The dash-dotted and dotted lines in the lower panels also represent the mean and maximum/minimum values, respectively. Note the maximum values for the alternative prey cycle is omitted for scale.

Figure S3: Bifurcation plot for the single-mimic system with morphological similarity. The upper panel illustrates the per capita harvest rate (i.e. the predation term in Equations 4 divided by the abundance of each corresponding prey) of the mimic (H_c, orange), the model (H_m, light blue), and the alternative prey (H_n, dark blue). The lower panel illustrates the predator's attack probability on the mimic (p_c, orange), the model (p_m, light blue) and the alternative prey (p_n, dark blue). Only the long-term mean is shown in the panels.

Figure S4: Effect of the model species value (v_m; x-axis) and the mimic–model similarity (k; y-axis) on the single Batesian mimicry system without abundance-dependent similarity. A lower model value indicates a lower profitability of consuming the model, which also influences the collective profitability of the mimic–model species pair; note the default value in the main text is $v_m = 0$. Different panels represent different variables: (a) mimic abundance (N_c), (b) model abundance (N_m), (c) predator abundance (N_p), (d) the attacked probability on the mimic (p_c), (e) the attacked probability on the model (p_m), and (f) the log(variance) of the predator abundance fluctuation, serving as an indicator of whether the system is cycling or not. See section Numerical simulations for parameter details.

Figure S5: Time series of species abundances in a four-mimicry system where predator recognition is based solely on mimic–model morphological similarity (k_{c_j}). The system includes: two Batesian mimics (N_{c_1} and N_{c_3}; orange and light orange, respectively) and two Müllerian mimics (N_{c_2} and N_{c_4}; green and light green, respectively), the model (N_m; light blue), the predator (N_p; red) and the alternative prey (N_n; dark blue). Note that the abundance of predators is divided by four for better visualization. Similarity (k_{c_j}) values as follows: $k_{c_1} = 0.4$, $k_{c_2} = 0.9$, $k_{c_3} = 0.5$ and $k_{c_4} = 0.6$. Other parameter follow the main text, with additional parameters for N_{c_3} and N_{c_4} including: $v_{c_3} = 2.5$, $v_{c_4} = 0$, $h_{c_3} = h_{c_4} = 1$, $r_{c_3} = r_{c_4} = 2$ and $K_{c_3} = K_{c_4} = 10$.

Figure S6: Effect of the Batesian mimic–model similarity (k_1; x-axis) and the Müllerian mimic–model similarity (k_2; y-axis) on the multi-mimicry system when recognition depends solely on morphological similarity (i.e. without abundance-dependent recognition). The different panels represent different variables: (a) Batesian mimic abundance (N_{c_1}), (b) attack probability on the Batesian mimic (p_{c_1}), (c) the Müllerian mimic abundance (N_{c_2}), (d) attack probability on the Müllerian mimic (p_{c_2}), (e) model abundance (N_m), (f) attack probability on the mimic (p_m), (g) the log(variance) of the predator fluctuations, serving as an indicator of community stability, and (h) the attack probability on the alternative prey (p_n). Note that the abundances of different state variables are on different scales to better show the pattern of each variable. See main text for other parameter values.

Figure S7: Effect of predator sensitivity to prey categories (σ; x-axis) and the mimic–model morphological similarity (k; y-axis) on the single Batesian mimicry system with abundance-dependent similarity. The different panels represent different variables: (a) Batesian mimic abundance (N_c), (b) attack probability on the Batesian mimic (p_c), (c) the model abundance (N_m), (d) attack probability on the model (p_m), (e) predator abundance (N_p), (f) the log(variance) of the predator abundance fluctuation as an indicator of system stability, (g) alternative prey abundance (N_n) and (h) the realized similarity due

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to abundance-dependent recognition (p_n). Note that the abundances of different state variables are on different scales to better show the pattern of each variable. See main text for other parameter values.

Appendix A: Equations for the single and the two-mimic system.

Appendix B: Analytical criteria for diet shifting.

Appendix C: Caution against multi-species signal detection theory.

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